

03 Assignment – Docker,

Docker Hub and Containerization

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1. L3 - Demonstrate Docker Compose using the Application Image and MySQL Image to start and stop all container services.

- Verify Docker & Docker Compose and change directory to project directory

```
Last login: Sun Feb  1 20:39:49 2026 from 27.57.41.157
[ec2-user@ip-172-31-15-198 ~]$ sudo su
[root@ip-172-31-15-198 ec2-user]# docker --version
Docker version 25.0.14, build 0bab007
[root@ip-172-31-15-198 ec2-user]# docker compose version
Docker Compose version v2.29.2
[root@ip-172-31-15-198 ec2-user]# docker compose version
Docker Compose version v2.29.2
[root@ip-172-31-15-198 ec2-user]# cd Docker_Assignment
[root@ip-172-31-15-198 Docker_Assignment]# ls
Dockerfile app.py requirements.txt
[root@ip-172-31-15-198 Docker_Assignment]#
```

- Create docker-compose.yml to handle the containers

```
version: "3.8"

services:
  app:
    image: image01:v1
    container_name: my-python-container
    ports:
      - "5000:5000"
    depends_on:
      - mysql
  mysql:
    image: mysql:8.0
    container_name: mysql_container
    environment:
      MYSQL_ROOT_PASSWORD: root
      MYSQL_DATABASE: appdb
      MYSQL_USER: appuser
      MYSQL_PASSWORD: app123
    ports:
      - "3306:3306"
```

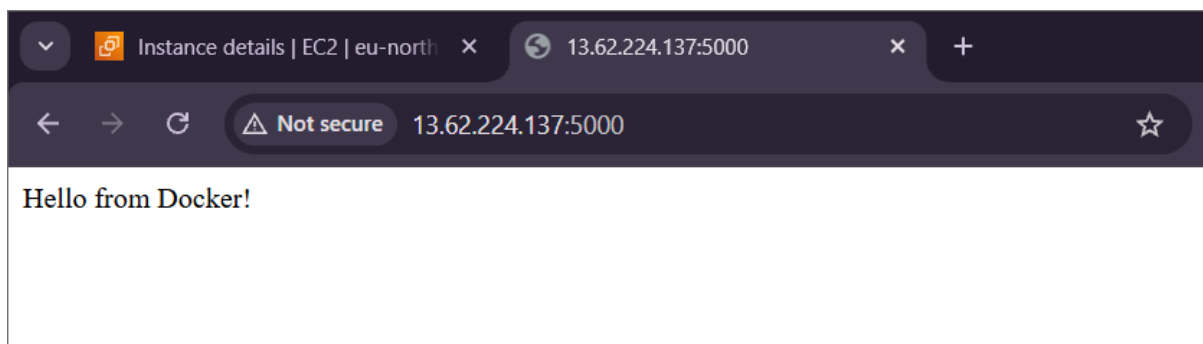
- Start All Container services

```
[root@ip-172-31-15-198 Docker_Assignment]# ls
Dockerfile  app.py  requirements.txt
[root@ip-172-31-15-198 Docker_Assignment]# vi docker-compose.yml
[root@ip-172-31-15-198 Docker_Assignment]# docker compose up -d
WARN[0000] /home/ec2-user/Docker_Assignment/docker-compose.yml: the attribute `version` is obsolete,
it will be ignored, please remove it to avoid potential confusion
[+] Running 12/12
  ✓ mysql Pulled                                17.8s
  ✓ c21bb7e51cd3 Pull complete                  3.3s
  ✓ d9bba75f9c49 Pull complete                  3.4s
  ✓ d7842cc74782 Pull complete                  3.4s
  ✓ ce1944d72bca Pull complete                  3.7s
  ✓ c21ebd7bbded Pull complete                  3.7s
  ✓ 142c5fc9ea8a Pull complete                  3.7s
  ✓ 526dbe6f3591 Pull complete                  5.3s
  ✓ b05b17de9630 Pull complete                  5.4s
  ✓ 3145e04eb470 Pull complete                 15.7s
  ✓ f79b7cf07833 Pull complete                 15.8s
  ✓ b4f9f94601d6 Pull complete                 15.8s
[+] Running 3/3
  ✓ Network docker_assignment_default Created    0.2s
  ✓ Container mysql_container Started           1.1s
  ✓ Container my_python_container Started        1.3s
[root@ip-172-31-15-198 Docker_Assignment]# docker compose ps
WARN[0000] /home/ec2-user/Docker_Assignment/docker-compose.yml: the attribute `version` is obsolete,
it will be ignored, please remove it to avoid potential confusion


| NAME                | IMAGE      | COMMAND                  | SERVICE | CREATED        | STATUS        |
|---------------------|------------|--------------------------|---------|----------------|---------------|
| my_python_container | image01:v1 | "python app.py"          | app     | 56 seconds ago | Up 55 seconds |
| mysql_container     | mysql:8.0  | "docker-entrypoint.s..." | mysql   | 56 seconds ago | Up 55 seconds |


[root@ip-172-31-15-198 Docker_Assignment]#
```

- Access application in browser



- Stop all container services

```
[root@ip-172-31-15-198 Docker_Assignment]# docker compose down
WARN[0000] /home/ec2-user/Docker_Assignment/docker-compose.yml: the attribute `version` is obsolete,
it will be ignored, please remove it to avoid potential confusion
[+] Running 3/3
  ✓ Container my_python_container Removed        10.7s
  ✓ Container mysql_container Removed            1.4s
  ✓ Network docker_assignment_default Removed    0.5s
[root@ip-172-31-15-198 Docker_Assignment]# docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
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